Compound assignment / short hand operators: Here we are using assignment operator with the combination of other operators as follows.

+=, -=, *=, %=, /=, <<=, >>=,….

Eg:

int a=10, b=20;

float c=5;

a+=15; i.e. a=a+15 ➔ a = 10 + 15 ➔ 25

b*=5; i.e. b=b*5 ➔ b = 20 * 5 = 100

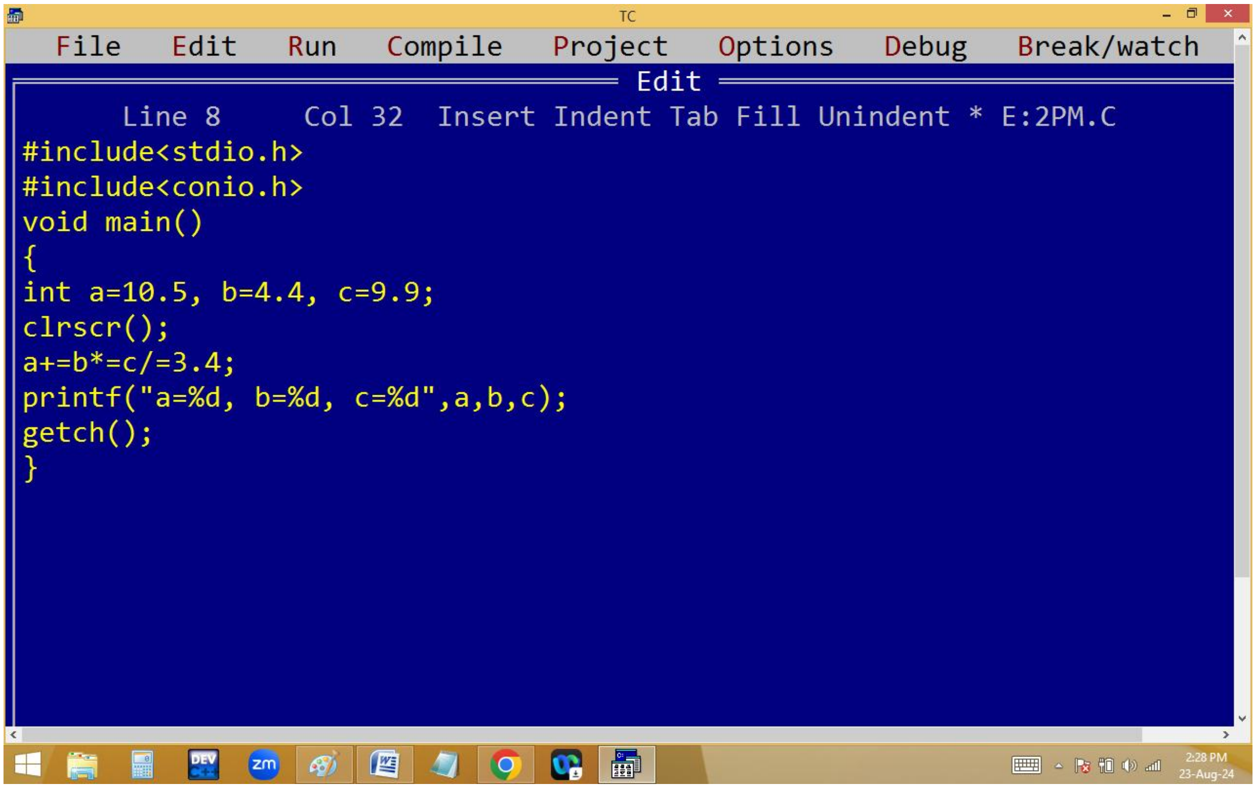
c/=2; i.e. c=c/2 ➔ c = 5/2 ➔ 2.500000

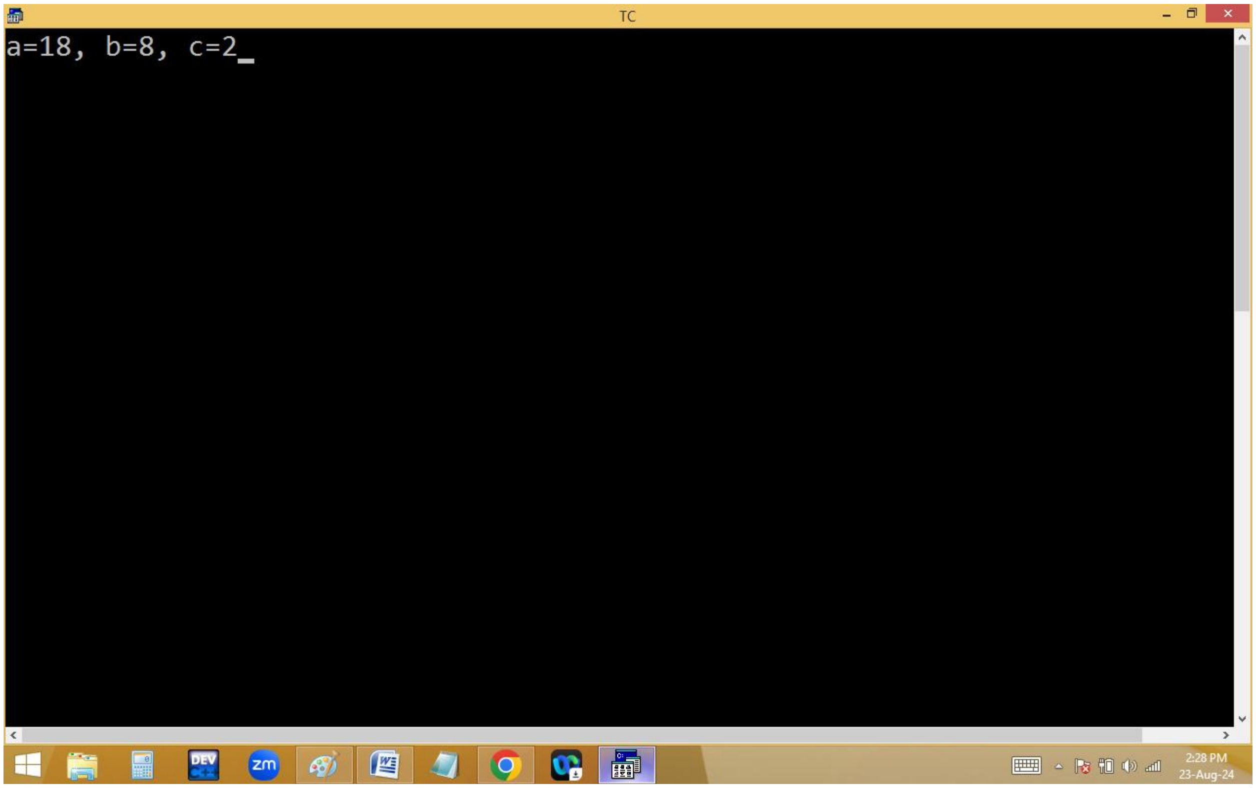
```
File   Edit   Run   Compile   Project   Options   Debug   Break/watch
      Line 8     Col 3    Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10, b=20; float c=5;
clrscr();
a+=15;
b*=5;
c/=2;
printf("a=%d, b=%d, c=%f\n",a, b, c);
getch();
}
```

```
a=25, b=100, c=2.500000
```

```
File    Edit    Run    Compile    Project    Options    Debug    Break/watch
      Line 7      Col 25   Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
a+=5+8; /* a=a+5+8=23 */_
printf("a=%d",a);
getch();
}
```

```
a=23
```


TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 8 Col 32 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10.5, b=4.4, c=9.9;
clrscr();
a+=b*=c/=3.4;
printf("a=%d, b=%d, c=%d",a,b,c);
getch();
}
2:28 PM
23-Aug-24



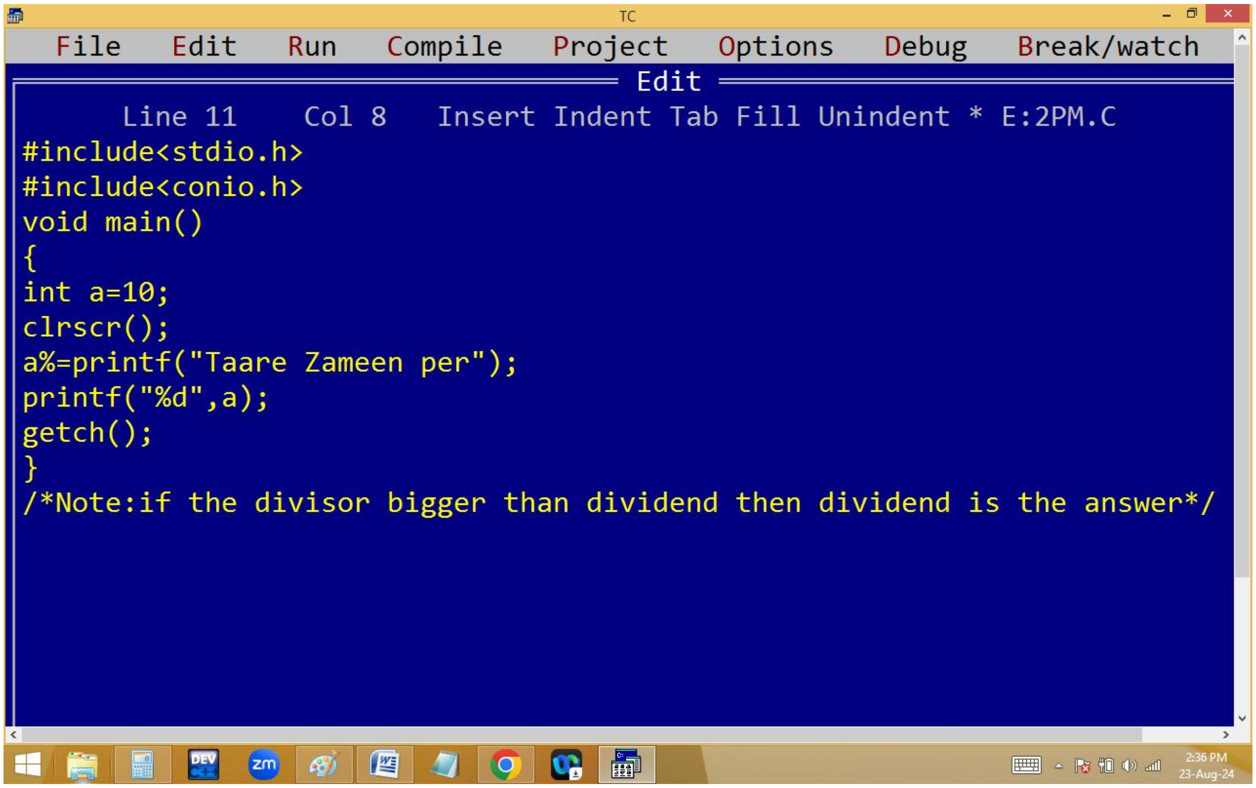
int a=10~~.5~~ b=4~~.4~~ c=9~~.9~~

a+= b*= c/=3.4;

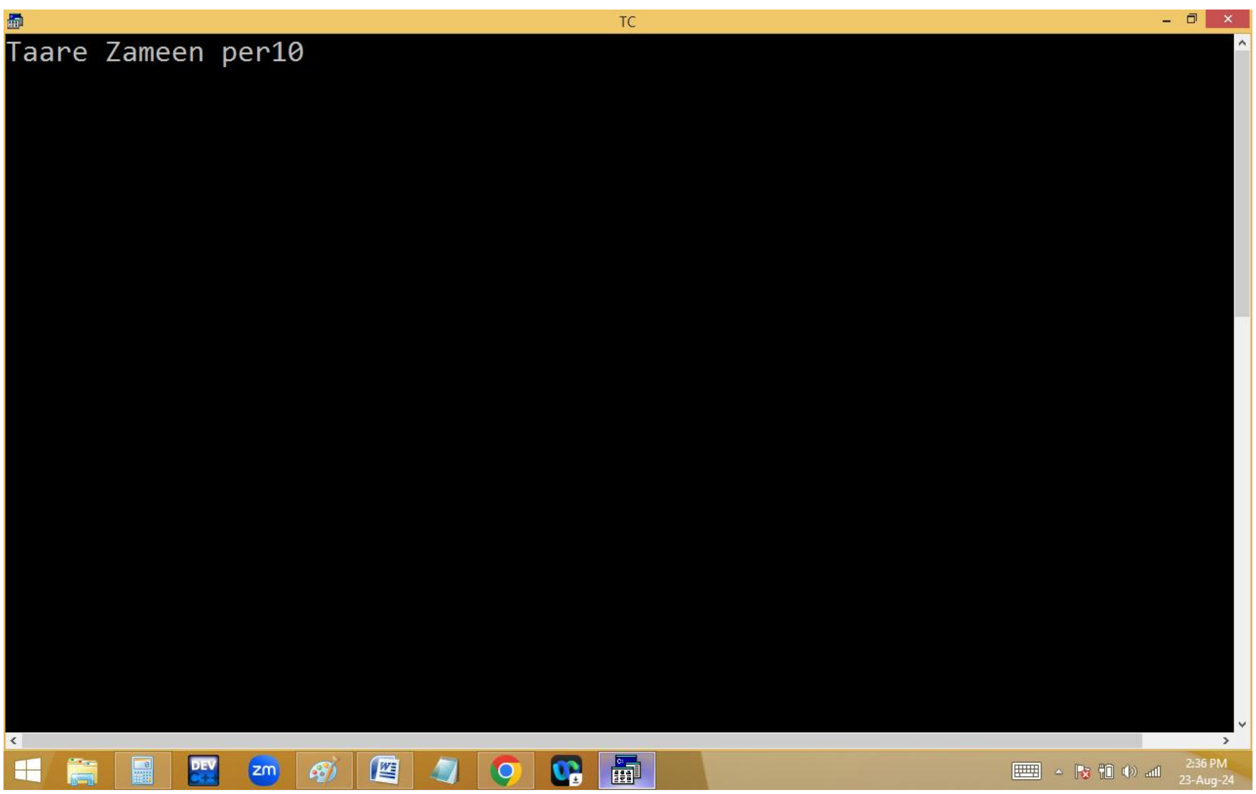
c = c / 3.4 ==> 9/3.4==> 2

b*=c ==> b = b*c ==> 4*2=8

a+=b ==> a=a+b ==> 10 + 8 = 18




```c
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
a%=printf("Taare Zameen per");
printf("%d",a);
getch();
}
/*Note:if the divisor bigger than dividend then dividend is the answer*/
```




```
Taare Zameen per10
```

Address operators:

1. & - Address of variable
2. * - address of another variable [pointer]

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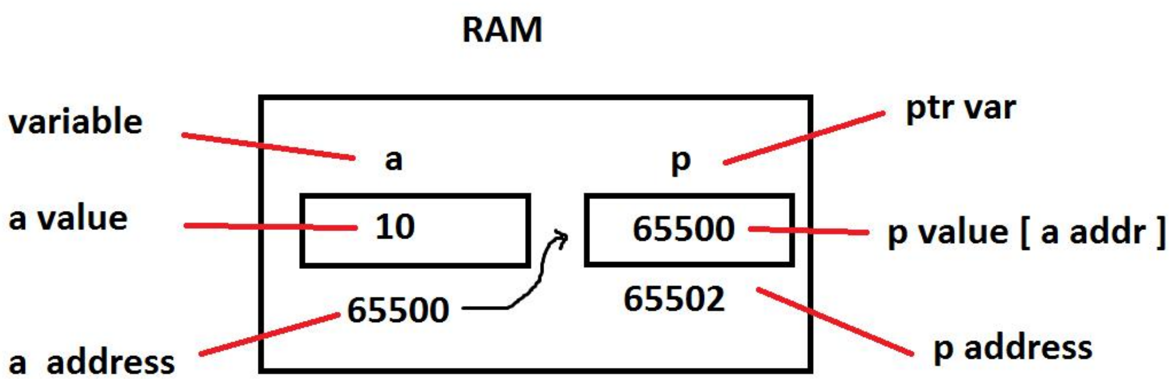
Edit

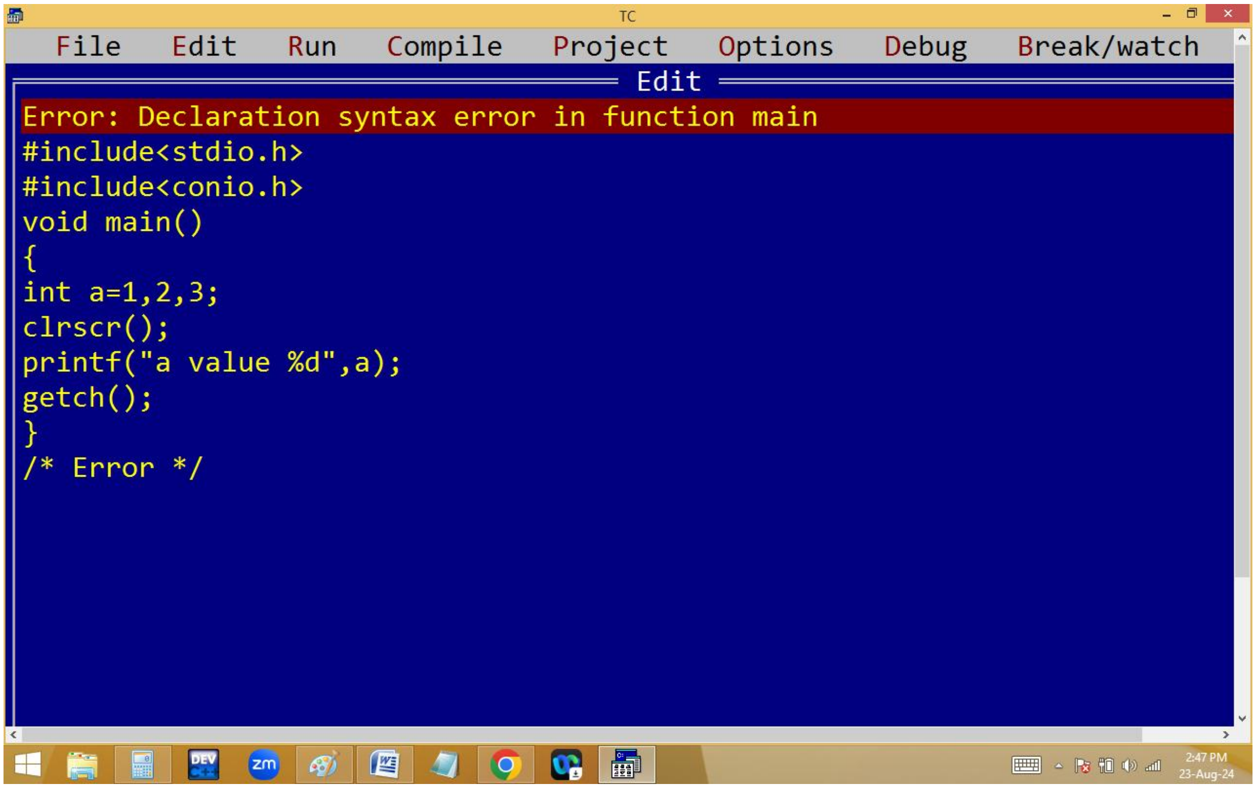
Line 7 Col 33 Insert Indent Tab Fill Unindent * E:2PM.C

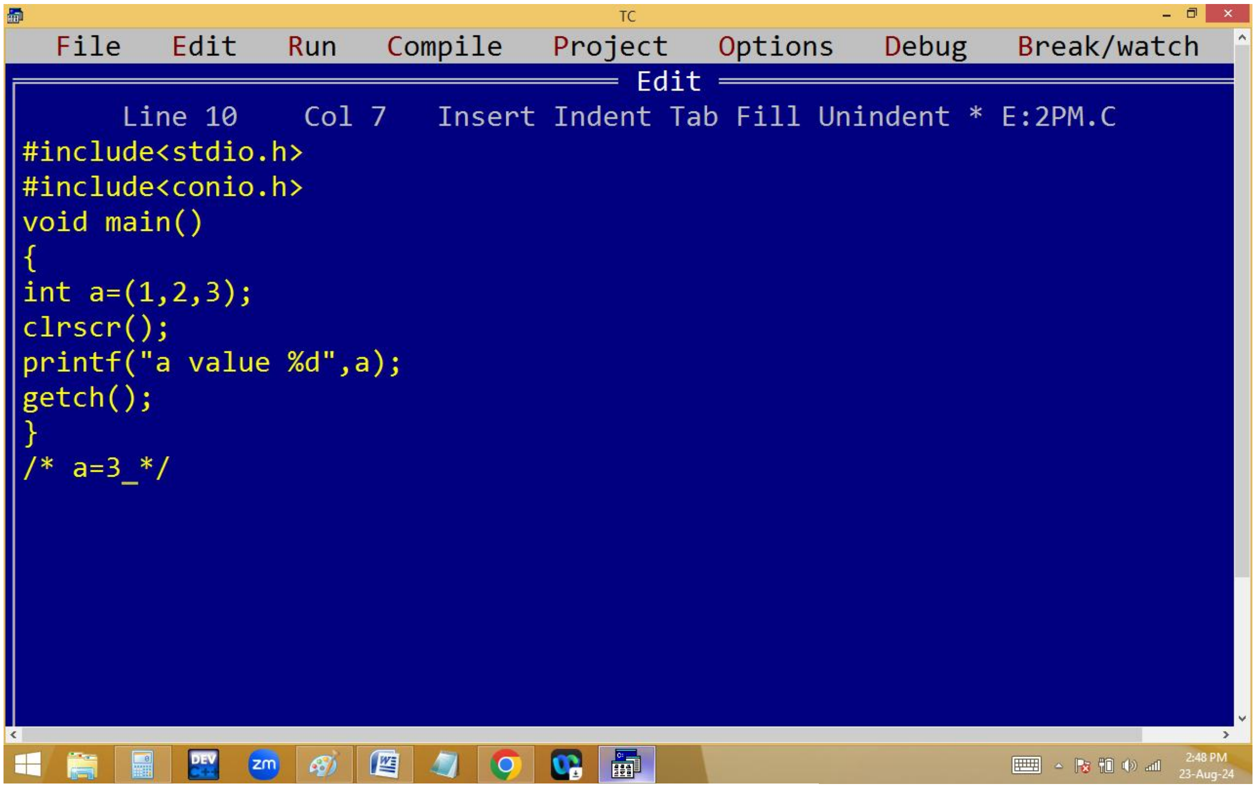
```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10; /* var */
int *p; /* pointer */
p = &a; /* a addr stored in p */_
clrscr();
printf("a value %d\n",a);
printf("a address %u\n",&a);
getch();
}
/*Note:if the divisor bigger than dividend then dividend is the answer*/
```

```
a value 10
a address 65500
```



( ) AND , SEPARATORS:


TC
File
Edit
Run
Compile
Project
Options
Debug
Break/watch
Edit
Error: Declaration syntax error in function main
#include<stdio.h>
#include<conio.h>
void main()
{
int a=1,2,3;
clrscr();
printf("a value %d",a);
getch();
}
/* Error */
2:47 PM
23-Aug-24



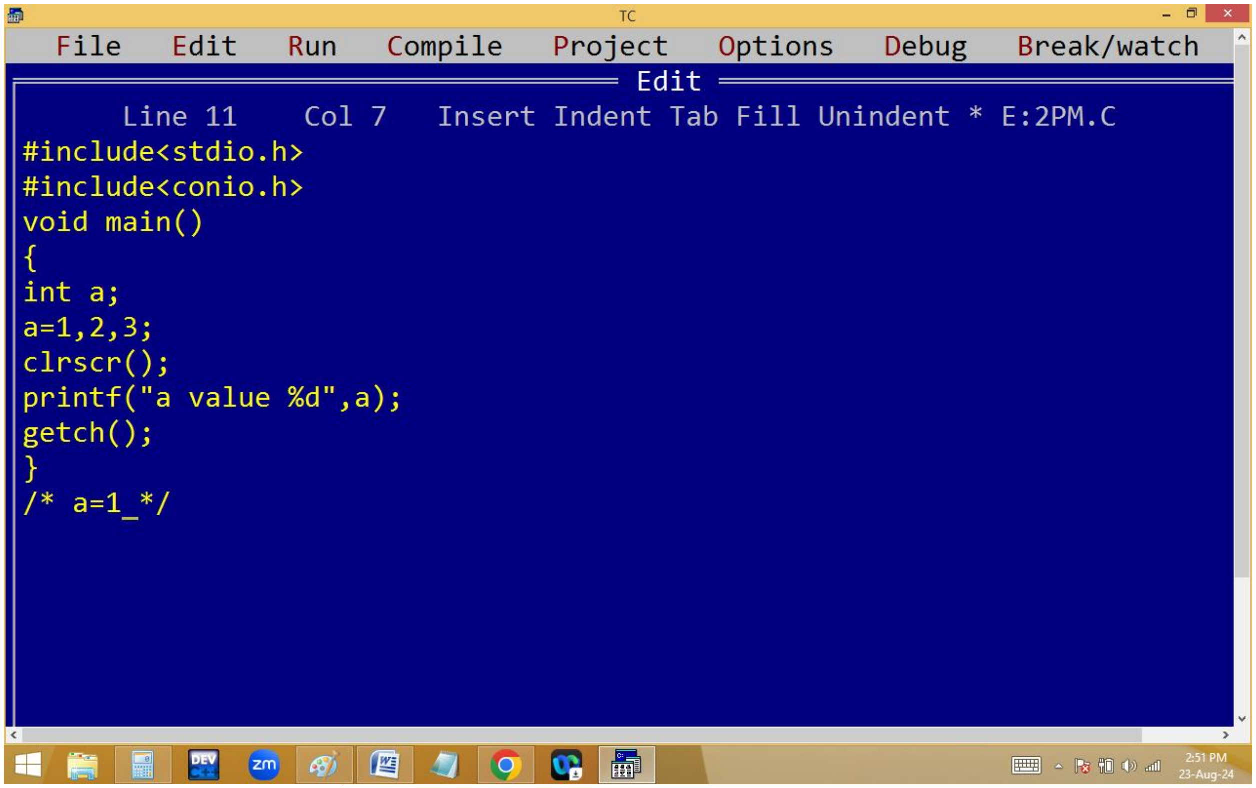
int a = (1, 2, 3);

L R

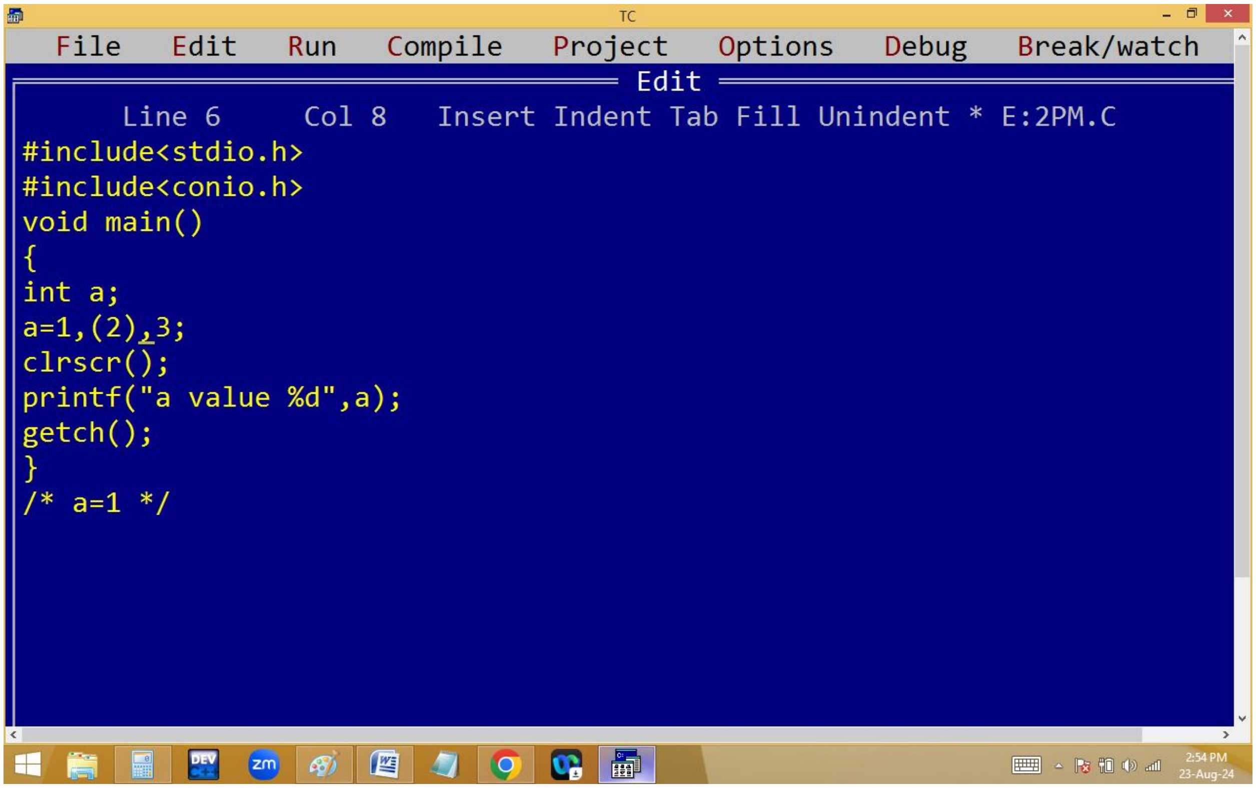
2 , 3

L R

3


TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 11 Col 7 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
a=1,2,3;
clrscr();
printf("a value %d",a);
getch();
}
/* a=1_*/
2:51 PM
23-Aug-24

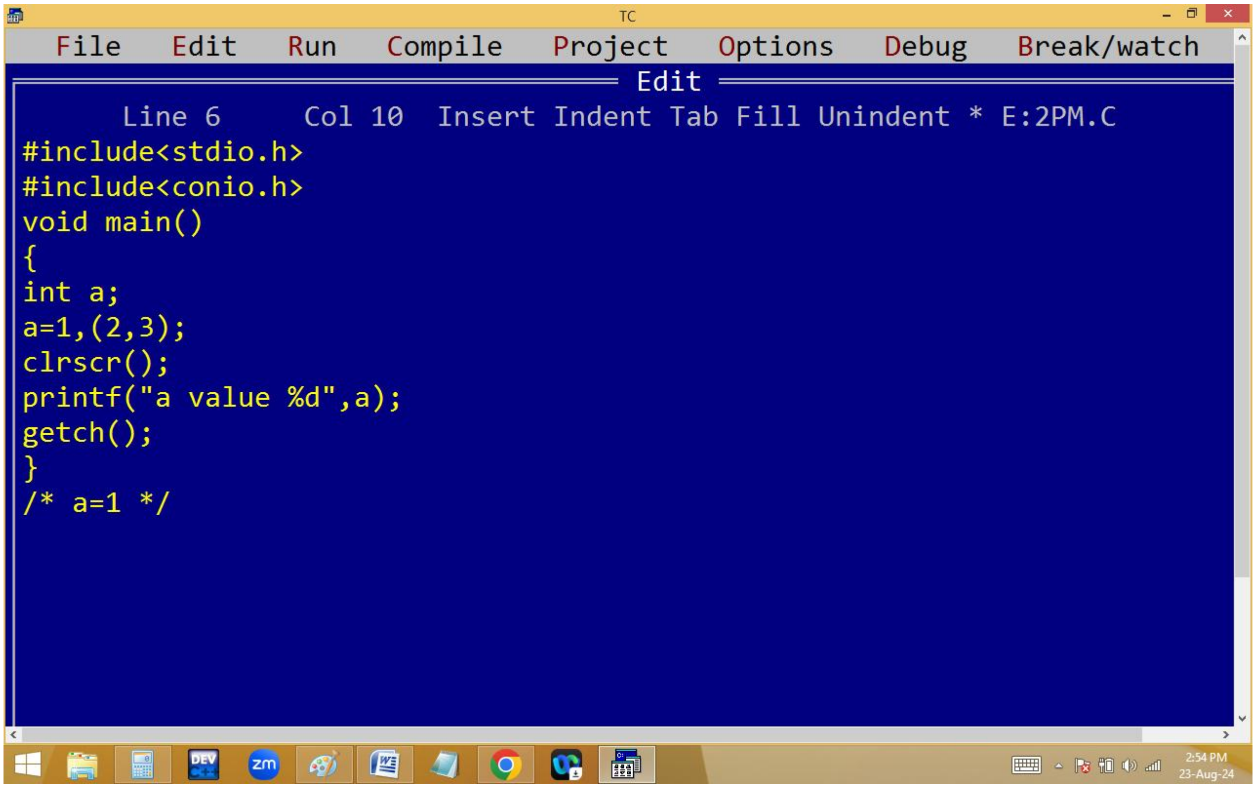
a=1, 2, 3;


TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 6 Col 8 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
a=1,(2),3;
clrscr();
printf("a value %d",a);
getch();
}
/* a=1 */
2:54 PM
23-Aug-24

a = 1 , (2), 3;

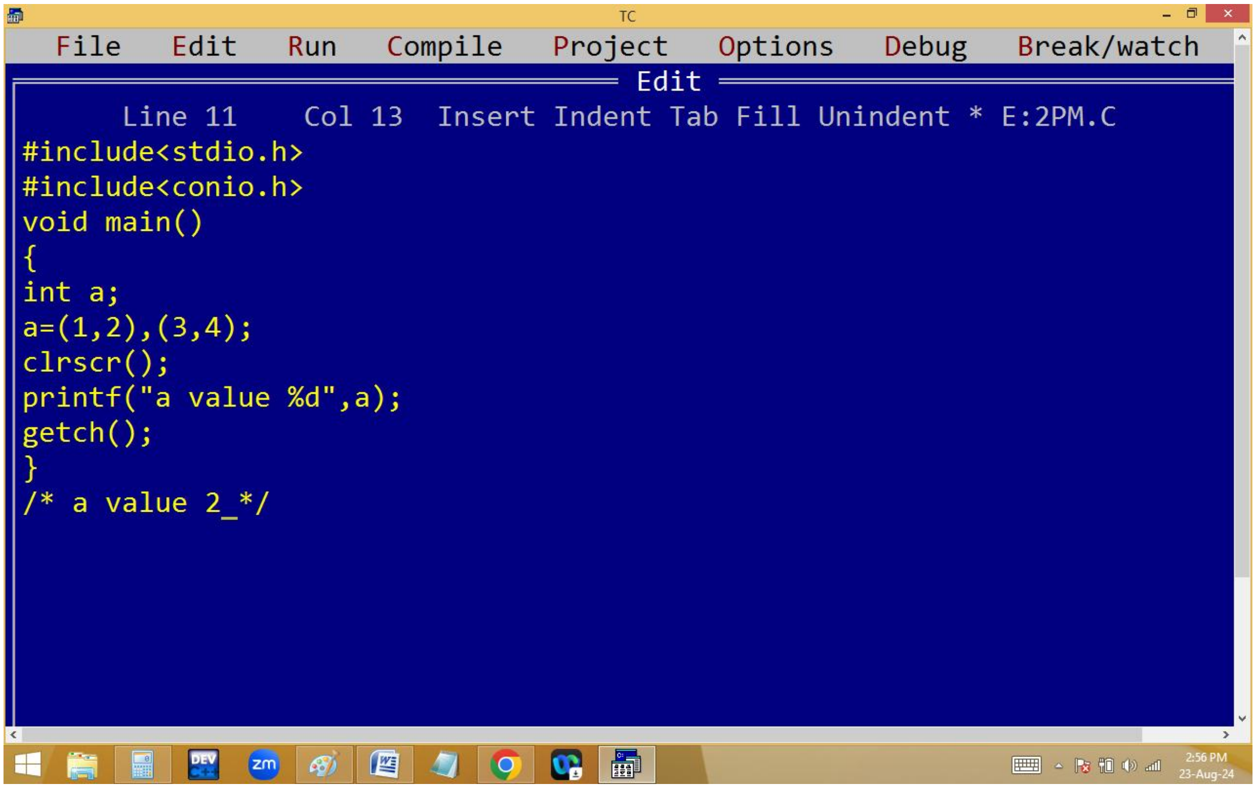
↓

a = 1 , 2 , 3 ;


TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 6 Col 10 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
a=1,(2,3);
clrscr();
printf("a value %d",a);
getch();
}
/* a=1 */
2:54 PM
23-Aug-24

a = 1, (2, 3);

a = 1 , 3


TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 11 Col 13 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
a=(1,2),(3,4);
clrscr();
printf("a value %d",a);
getch();
}
/* a value 2_*/
2:56 PM
23-Aug-24

a = (1,2) , (3,4);

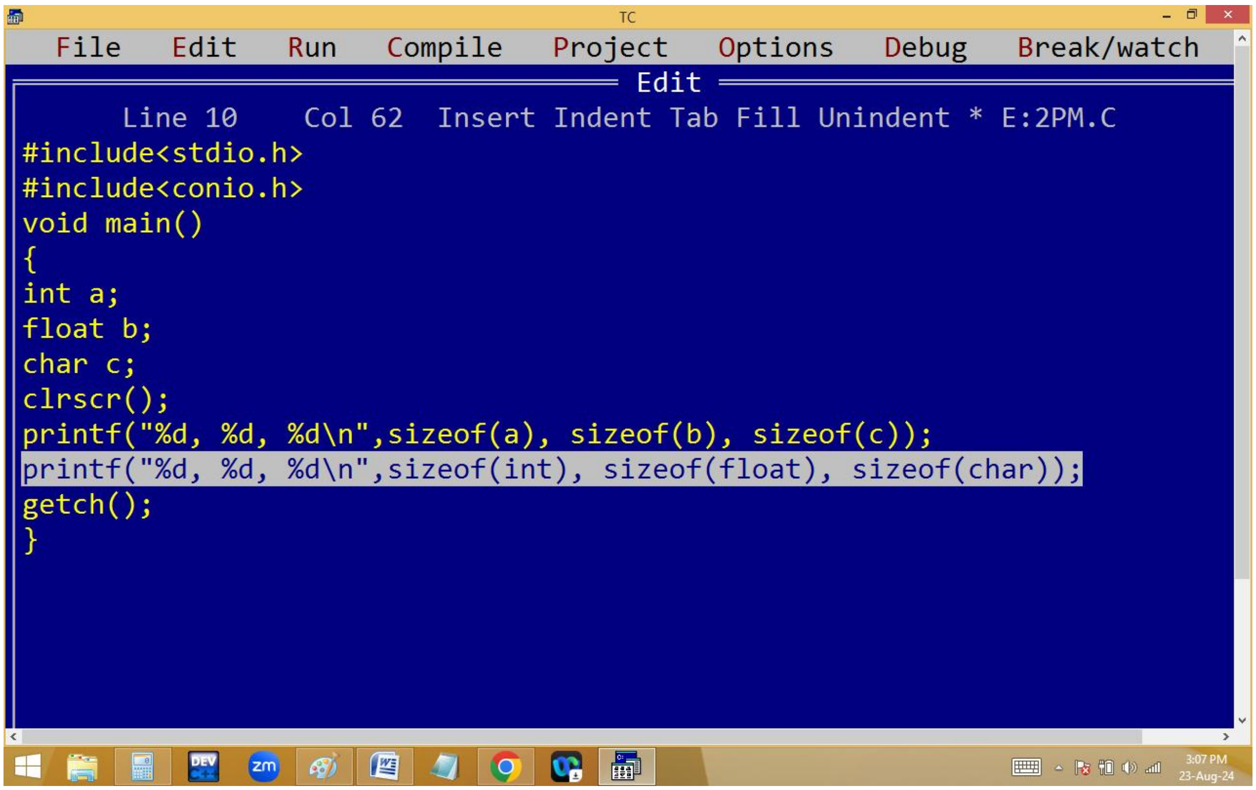
a= 2 , 4

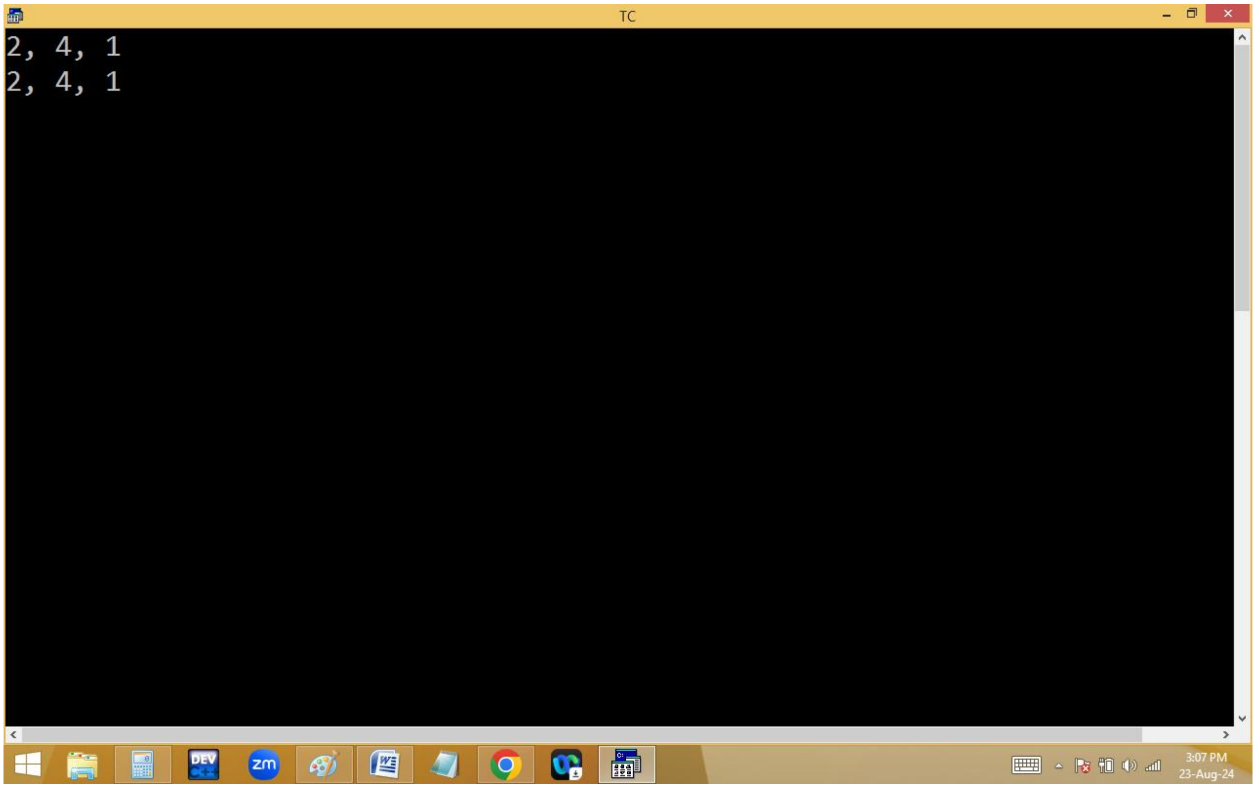
```
File   Edit   Run   Compile   Project   Options   Debug   Break/watch
                                 Edit
      Line 11     Col 1   Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
clrscr();
a= printf("Dilse"), printf("Kahona Pyar Hai");
printf("\na value %d",a);
getch();
}
```

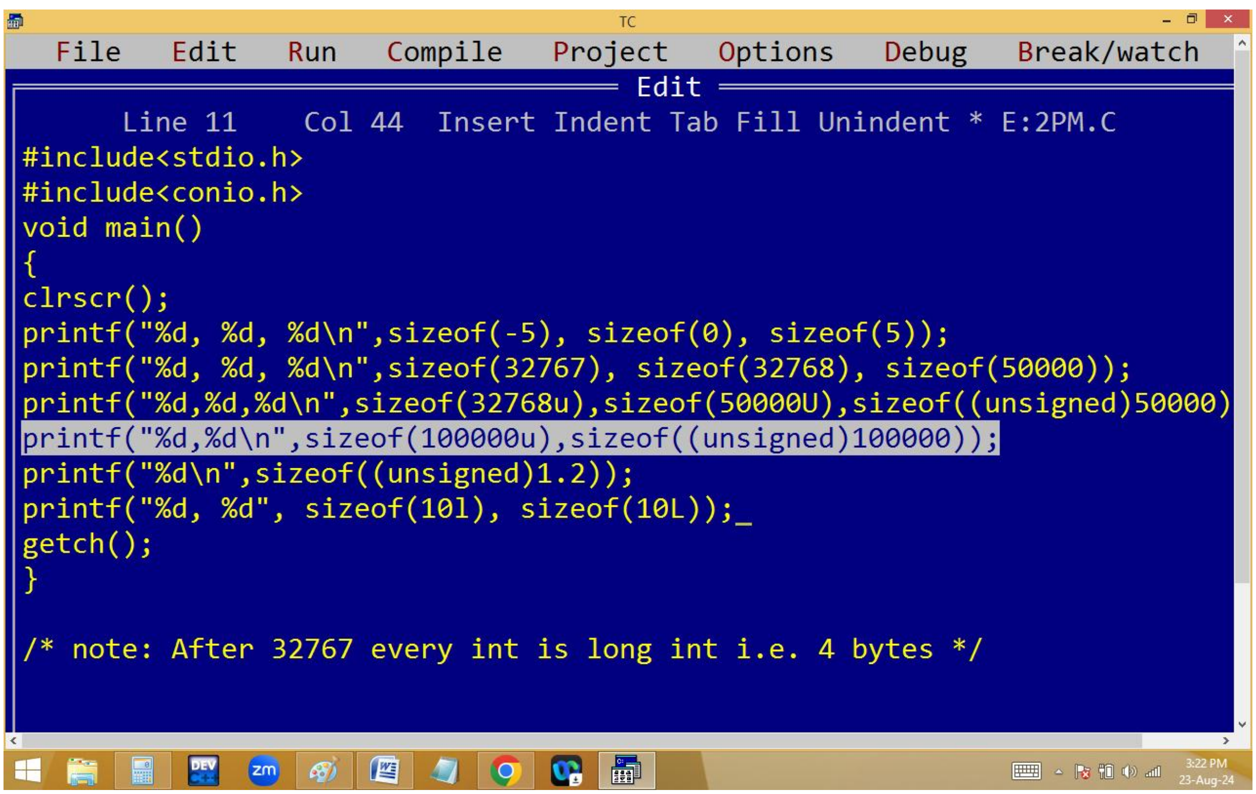
```
DilseKahona Pyar Hai
a value 5
```

sizeof():

It returns the no of bytes taken by a variable / data type / value.




TC
2, 4, 1
2, 4, 1
3:07 PM
23-Aug-24


TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 11 Col 44 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d, %d, %d\n",sizeof(-5), sizeof(0), sizeof(5));
printf("%d, %d, %d\n",sizeof(32767), sizeof(32768), sizeof(50000));
printf("%d,%d,%d\n",sizeof(32768u),sizeof(50000U),sizeof((unsigned)50000)
printf("%d,%d\n",sizeof(100000u),sizeof((unsigned)100000));
printf("%d\n",sizeof((unsigned)1.2));
printf("%d, %d", sizeof(10l), sizeof(10L));_
getch();
}
/* note: After 32767 every int is long int i.e. 4 bytes */
3:22 PM
23-Aug-24

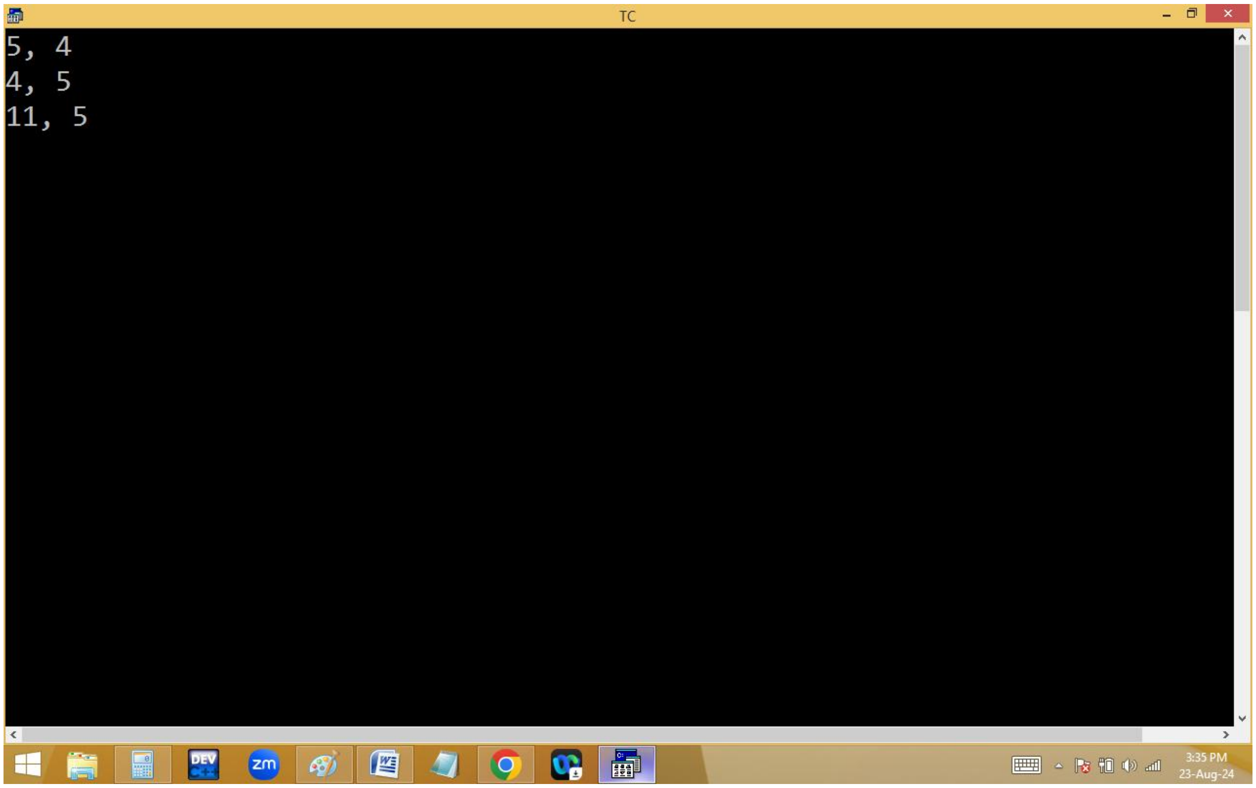
```
2, 2, 2
2, 4, 4
2,2,2
4,2
2
4, 4
```

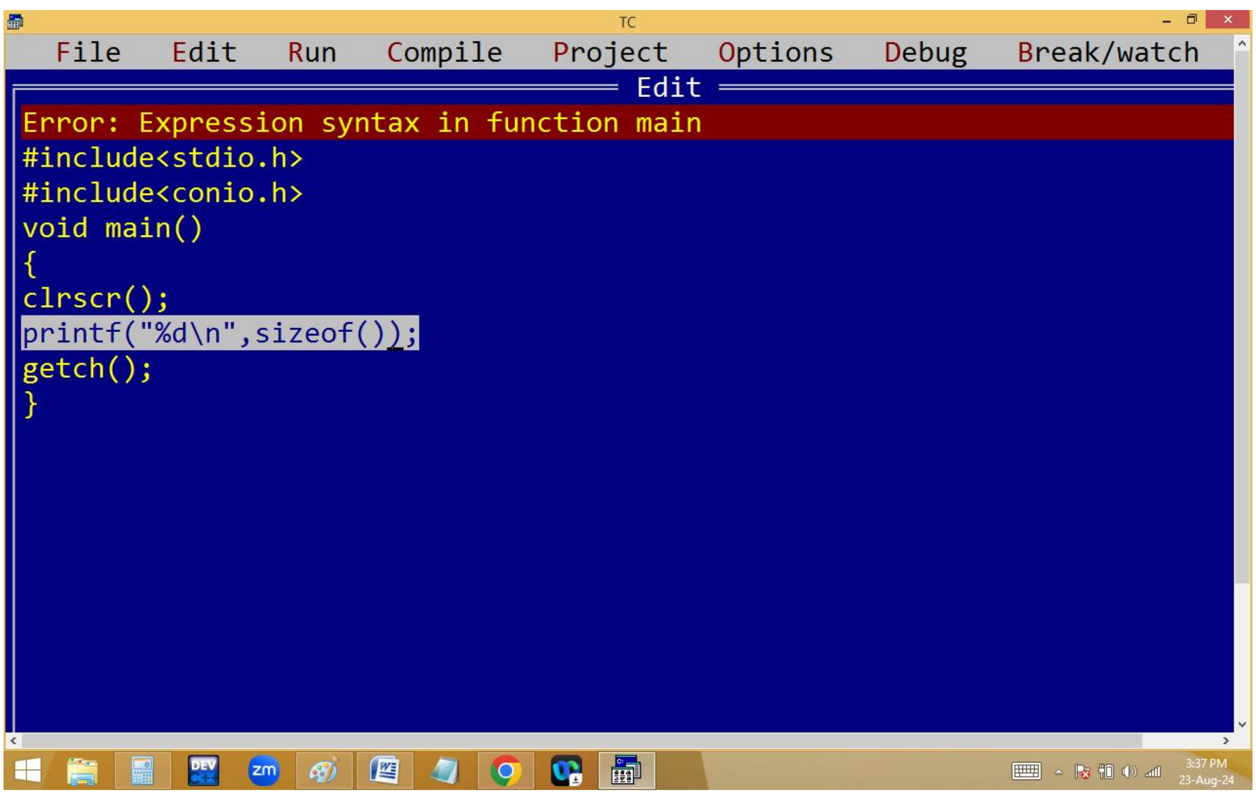
```
 File   Edit   Run   Compile   Project   Options   Debug   Break/watch
                                Edit
      Line 1      Col 51  Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
float a=1.1;
clrscr();
printf("%d, %d\n",sizeof(a),sizeof(1.1));
printf("%d, %d\n",sizeof(1.1f),sizeof((float)1.1));
printf("%d, %d\n",sizeof(1.1l),sizeof(1.1L));
printf("%d, %d\n",sizeof(10f),sizeof((long double)11));
getch();
}

/*note:floating value without float data type is a double i.e. 8 bytes */
```

```
4, 8
4, 4
10, 10
4, 10
```

```
File  Edit  Run  Compile  Project  Options  Debug  Break/watch
Edit
Line 10   Col 1   Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s1[5]="Hyd", s2[]="Hyd";
clrscr();
printf("%d, %d\n",sizeof(s1),sizeof(s2));
printf("%d, %d\n",sizeof("Hyd"),sizeof("Hyd")+1);
printf("%d, %d\n",sizeof("Hyd 500016"),sizeof("1.23"));
getch();
}
```


TC
5, 4
4, 5
11, 5
3:35 PM
23-Aug-24


TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Error: Expression syntax in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",sizeof());
getch();
}
3:37 PM
23-Aug-24

TC

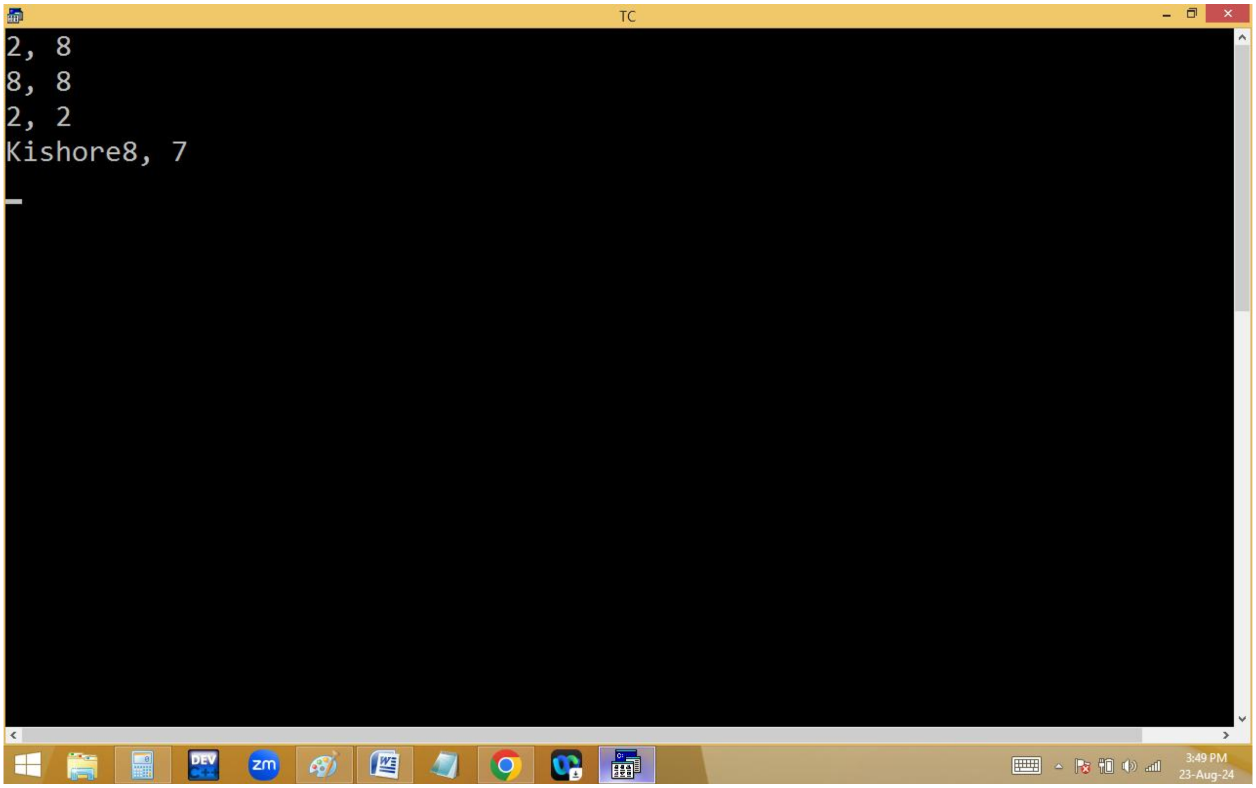
File Edit Run Compile Project Options Debug Break/watch

Edit

Line 9 Col 35 Insert Indent Tab Fill Unindent * E:2PM.C

```c
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d, %d\n",sizeof(1.2,10),sizeof(10,1.2));
printf("%d, %d\n",sizeof(1.2+10),sizeof(10+1.2));
printf("%d, %d\n",sizeof(sizeof("Kishore")),sizeof(printf("Kishore")));
printf("%d, %d\n",sizeof("Kishore"),printf("Kishore"));
getch();
}
```

3:49 PM
23-Aug-24



sizeof(sizeof("Kishore"))

8

int

2

sizeof(printf("Kishore"))

7

int

2

p("%d, %d", sizeof("kishore"), printf("kishore"));

8 , 7

Kishore 8, 7